Data Communication and Computer Networks

Chapter 1 | Data Communication Fundamentals

Unit 1 - Data communication, networks, OSI and TCP/IP, signals, line coding, PCM (8 hrs)

Syllabus covered

Overview of data communication and its components; fundamentals of networks, network criteria, topologies and categories; protocols and standards; reference models - OSI and TCP/IP; physical layer - analog and digital signals, data rate limits, transmission impairments; digital transmission - line coding, sampling, transmission modes.

What is inside

Section	Question type	Marks	Count
Section A	Short answer - definitions, one-liners, small numericals	2	52
Section B	Long answer - full explanations with tables and diagrams	7	14

How to use this bank

- Read Section A first: these are the definitions and small numericals that appear in Part A of every paper (2 marks each).
- Then take one 7-mark question a day from Section B and write it out from memory using only the bold headings.
- Every answer is written as short, keyword-first points. In the exam, one bullet is roughly one mark-bearing idea.
- Numbers worth memorising are repeated inside the answers: 2 x B x log2 L, C = B log2 (1 + SNR), -3 dB for half power, 64 kbps for one PCM voice channel.

Source tags

- **PYQ <year>** - asked in a C. V. Raman Global University end-semester paper (2022, 2023, 2024-25 or the 2024-25 backlog paper).
- **Model** - from the printed unit-wise model question paper.
- **Assignment** - from the course assignment for this unit.
- **Notes** - from the lecture slides, not yet asked but examinable.

Section A | 2-mark questions

Answer in two to four lines. Each point below is one mark-bearing idea, so a single line written in the exam earns the mark.

A1. Define data communication. [Notes]

- **Exchange of data** between two devices through some form of transmission medium, such as a wire cable.
- **Telecommunication** = communication at a distance. **Data** = information in whatever form the two parties agree upon.
- **Test of a good system**: the data must reach the right user, unchanged, on time, with low jitter.

A2. Write the different components of a data communication system. [PYQ 2023]

- **Message** - the information: text, numbers, pictures, audio, video.
- **Sender** - the device that sends the message (computer, telephone handset, video camera).
- **Receiver** - the device that receives it (computer, television).
- **Transmission medium** - the physical path: twisted pair, coaxial, fiber-optic cable, radio waves.
- **Protocol** - the rules of the conversation. Without it two connected devices are connected but not communicating.

A3. State the four fundamental characteristics of an effective data communication system. [Notes]

- **Delivery** - data reach the correct destination, and only the intended user.
- **Accuracy** - data arrive unchanged; changed and uncorrected data are unusable.
- **Timeliness** - data arrive in time, in the same order, with no delay for real-time audio and video.
- **Jitter** - the variation in packet arrival time must be small; large jitter shows as uneven video.

Memory hook - D-A-T-J: **D**eliver, **A**ccurately, **T**imely, without **J**itter.

A4. Define jitter. [Notes]

- Jitter is the **variation in the arrival time** of packets of the same stream.
- Packets spaced unevenly at the receiver give uneven audio and video quality.
- It is separate from delay: a stream may be uniformly late and still be usable, but it cannot be irregularly late.

A5. Illustrate protocol and its requirement in data communication. Give an example. [PYQ 2022; PYQ 2024-25 backlog]

- A **protocol is a set of rules** that governs data communication: what to communicate, when, and how.
- Three key elements - **syntax** (format or structure of the data), **semantics** (meaning of each field), **timing** (when to send and how fast).
- **Requirement**: hardware only provides a path. Without agreed rules the receiver cannot interpret the bits, cannot detect errors and cannot pace the sender.
- **Example**: TCP/IP, HTTP, or the simple rule that a frame starts with the flag 01111110.

A6. Write the principles of protocol layering and explain logical connections. [PYQ 2024-25 backlog]

1. **One job per layer**: each layer performs a well-defined, self-contained function.
2. **Service without knowledge**: a layer uses the services of the layer below and offers services to the layer above, without knowing how those services are implemented.
3. **Peer-to-peer logical connection**: layer n on the sender appears to talk directly to layer n on the receiver. The logical path is horizontal; the actual data path is vertical down the stack, across the link and up the stack.

- **Benefit**: modularity, easier design and debugging, and interoperability between vendors - the reason OSI and TCP/IP are layered.
- **Cost**: a header at every layer, so overhead grows with the number of layers.

A7. Explain the concept of encapsulation and decapsulation with a diagram. [PYQ 2024-25 backlog]

- **Encapsulation**: as data moves *down* the sender's stack, each layer adds its own header (and the data link layer a trailer).
- **Order**: data, then segment (transport header), packet (network header), frame (data link header and trailer), then bits.
- **De-encapsulation**: at the receiver each layer removes exactly the header its peer added, and passes the rest upward.
- **Diagram to draw**: two vertical stacks with horizontal arrows at each layer, and the packet growing a header at every step down.

Memory hook - the envelope idea: you write a letter, put it in an envelope (header), put the envelope in a mailbag (trailer plus frame), and the receiver reverses the order.

A8. Differentiate half-duplex and full-duplex transmission modes with examples. [Model]

Feature	Half-duplex	Full-duplex
Direction	Both ways, but not at the same time	Both ways at the same time
Channel capacity	Entire capacity taken by the transmitting device	Capacity divided between the two directions
Turn taking	Required	Not required
Example	Walkie-talkie	Telephone network

A9. Compare and contrast parallel and serial transmission mode. [PYQ 2023; Model]

Feature	Parallel	Serial
Wires	n wires carry n bits at one clock tick	One channel, one bit after another
Speed	High, for the same clock	Lower, but run at a much higher clock
Cost	High - n channels needed	Low - one channel
Distance	Short, because of skew between wires	Long
Use	Computer to printer, internal buses	Modems, LANs, long-haul links

A10. Differentiate asynchronous and synchronous transmission. [Notes]

Feature	Asynchronous	Synchronous
Framing	1 start bit (0) and 1 or more stop bits (1) per byte	No start or stop bits, no gaps
Gap between bytes	Allowed	Not allowed
Who groups the bits	The start and stop bits do it	The receiver does it
Overhead	High, about 20 per cent of capacity	Low
Use	Keyboard, low-speed terminals	Bulk data, high-speed links

A11. Mention the four basic network topologies and cite an advantage of each type. [PYQ 2022; Model]

- **Mesh** - every device linked to every other: most robust, a failed link disables only itself.
- **Star** - each device linked to a central hub: easy fault identification, a failed link affects one device only.
- **Bus** - one backbone cable with taps: least cabling of all.
- **Ring** - each device linked to its two neighbours: a device can be added or removed by changing only two connections.

A12. Discuss the principle of data communication followed in ring topology. [PYQ 2023]

- Each device has a **dedicated point-to-point link to the two devices on either side**; for n devices there are n links.
- Every device contains a **repeater**: it regenerates the signal before passing it on, so the ring can cover a larger area.
- Data travel in **one direction** (unidirectional), station to station, until they reach the destination.
- Access is controlled - a station waits for a free token or its turn - so there are no collisions.
- Weakness: a break in the ring stops the whole network unless a dual ring is used.

A13. For n devices in a network, find the number of cable links required for mesh, ring and star topology. [Model]

- **Mesh**: n(n-1)/2 duplex links, because each of n devices needs a link to the other n-1, and each link is counted twice. Ports = n(n-1).
- **Ring**: n links, one from each device to its neighbour.
- **Star**: n links, one from each device to the hub.
- **Check**: for n = 8, mesh = 28, ring = 8, star = 8.

A14. Assume 8 devices are arranged in a mesh topology, all using simplex mode. Calculate the number of ports in each device and the number of cables required. [PYQ 2024-25 backlog]

1. In simplex mode a link carries data in one direction only, so every pair of devices needs **two** links, one each way.
2. Cables = n(n-1) = 8 x 7 = **56 cables**.
3. Ports in each device = n-1 = **7 ports**, so 8 x 7 = 56 ports in all.
4. In duplex mode the same network would need only 28 cables and 56 ports.

A15. Define network, node and link. [Notes]

- **Network** - a set of devices connected by communication links.
- **Node** - any device that can send or receive data generated by other nodes: computer, printer, switch.
- **Link** - the communication pathway that carries data from one device to another: cable, optical fiber, free space.

A16. State the three network criteria and say how each is measured. [Notes]

- **Performance** - measured by transit time, response time, throughput and delay.
- **Reliability** - measured by the frequency of failure, the recovery time, and robustness in a catastrophe.
- **Security** - protection of data from unauthorised access, plus the policies and procedures used.

A17. Define LAN, MAN and WAN in one line each, with one example. [Notes]

- **LAN** - privately owned, connects hosts in one office, building or campus (usually under 2 miles); example: college campus network.
- **MAN** - covers a town or city, between LAN and WAN in size; example: cable TV network providing Internet.
- **WAN** - long-distance transmission across a country or continent, interconnects LANs and MANs; example: the Internet.

A18. What are the principal causes of transmission impairments? [PYQ 2023; PYQ 2024-25 backlog]

- **Attenuation** - loss of energy with distance, measured in dB.
- **Distortion** - change of shape, because each frequency component travels at a different speed and arrives with a different delay.
- **Noise** - unwanted signal added on the way: thermal (electrons), induced (motors, crosstalk), impulse (lightning).
- **Result**: the signal at the end of the medium is not the signal that entered it.

A19. A signal travels from point X to point Y. At X the power is 300 W and at Y it is 250 W. Determine the attenuation in decibels. [PYQ 2022]

1. dB = 10 log10(P2/P1) = 10 log10(250/300).
2. 250/300 = 0.8333, log100.8333 = -0.0792.
3. **Attenuation = -0.79 dB** (a loss of about 0.8 dB).

A20. A signal travels through a medium and its power is reduced to one-half. Calculate the attenuation. [PYQ 2024-25 backlog]

1. dB = 10 log10(0.5) = 10 x (-0.3010).
2. **Attenuation = -3.01 dB, about -3 dB.**
3. Half power is always -3 dB; this is the single most useful figure to remember.

A21. A signal passes through an amplifier and its power is increased 10 times. Find the gain. [Model]

1. Gain = 10 log10(10) = **10 dB**.
2. Rule to remember: 10 times power = 10 dB, 2 times = 3 dB, half = -3 dB.

A22. Illustrate the significance of signal-to-noise ratio in data communication. [PYQ 2024-25]

- **SNR = Psignal/Pnoise**; in decibels SNRdB = 10 log10 SNR.
- It fixes the **capacity** of a real channel: C = B log2(1 + SNR). A higher SNR allows a higher bit rate.
- It fixes how many **signal levels** can be distinguished; noise sets the smallest reliable spacing between levels.
- It is the quantity a **link budget** is built on - repeaters, shielding and twisting all exist to keep SNR high.

A23. A channel has a bandwidth of 50 MHz. Determine the channel capacity for a signal-to-noise ratio of 70. [PYQ 2022]

1. C = B log2(1 + SNR) = 50 x 10^6 x log271.
2. log271 = 6.1497.
3. **C = 307.5 Mbps.**

A24. Distinguish between data rate and signal rate. [PYQ 2024-25 backlog]

- **Data rate (bit rate, N)** - number of data elements (bits) sent per second, in bps.
- **Signal rate (baud rate, S)** - number of signal elements sent per second, in baud.
- **Relation**: S = c x N x 1/r, where r = data elements per signal element and c = case factor (0 to 1, average 1/2).
- **Bandwidth**: Bmin = c x N x 1/r, since bandwidth follows the signal rate, not the bit rate.

A25. Distinguish between a signal element and a data element. [Model]

- **Data element** - the smallest entity that carries information: one bit. It is what we *need* to send.
- **Signal element** - the shortest unit of the digital signal in time. It is what we *can* send.
- One signal element may carry several data elements: with L levels it carries log2L bits.

A26. An analog signal has a frequency component of 15 kHz and is sampled at 20 kHz. Compute the sampling frequency and comment. [PYQ 2024-25]

1. The sampling frequency given is **fs = 20 kHz**.
2. The sampling theorem requires fs at least 2 x fmax = 2 x 15 = **30 kHz**.
3. **Comment**: 20 kHz is below the limit, so the signal is **aliased** - the reconstructed wave is a false lower-frequency signal. The rate must be raised to at least 30 kHz.

A27. State the sampling theorem. [Notes]

- fs must be at least **2 x fmax**, twice the highest frequency in the signal.
- Reason: below that rate, samples cannot tell the original wave from a lower-frequency wave passing through the same points.
- A low-pass signal of bandwidth 200 kHz therefore needs 400,000 samples per second.

A28. What is quantization error and how can it be reduced? [Model]

- **Quantization** rounds each sample to the nearest of L levels.
- **Quantization error** = actual sample value minus the quantized value; it is heard as quantization noise.
- **Reduced** by increasing the number of levels - every extra bit per sample halves the step - or by non-uniform quantization, which uses small steps for small signals.

A29. Write the different steps of pulse code modulation. [Model]

1. **Sampling** - measure the analog signal every Ts seconds at fs at least 2 fmax (PAM pulses).
2. **Quantization** - round each sample to the nearest of L levels.
3. **Encoding** - replace each level by its n-bit code, where L = 2^n.
4. Bit rate = fs x n; for telephony 8000 x 8 = 64 kbps.

A30. Compare PCM and delta modulation. [Model]

Feature	PCM	Delta modulation
Bits per sample	n bits (several)	1 bit
What is sent	The quantized sample value	Only the direction of change
Bit rate and bandwidth	High	Low
Hardware	Complex	Simple
Accuracy	Good	Slope-overload and granular noise

A31. What is the advantage of delta modulation over PCM? [Model]

- It sends **one bit per sample**, so the bit rate and bandwidth requirement are far lower.
- The circuitry is much **simpler** - a comparator and an up/down counter.
- It suits speech and low-cost coders where a small loss of quality is acceptable.

A32. Between RZ and NRZ encoding, which requires higher bandwidth and why? [Model]

- **RZ needs higher bandwidth.**
- Every bit returns to zero in the middle, so there are **two transitions per bit** instead of at most one.
- More transitions per second means a higher signal rate for the same data rate, and bandwidth follows the signal rate.
- The reward is self-clocking and no DC component.

A33. What does the Nyquist theorem have to do with communications? [Model]

- It gives the **maximum bit rate of a noiseless channel**: bit rate = 2 x B x log2L.
- It says a signal of bandwidth B can be sampled 2B times per second, and each sample carries log2L bits.
- It sets the ceiling that no modulation can exceed when there is no noise.

A34. What does the Shannon capacity have to do with communications? [Model]

- It gives the **maximum error-free bit rate of a noisy channel**: C = B log2(1 + SNR).
- No encoding scheme can beat it; a telephone line of 3000 Hz and SNR 3162 gives about 34.88 kbps.
- To send faster, widen the bandwidth or improve the SNR.

A35. A noiseless channel with a bandwidth of 6000 Hz is used for transmitting a signal with four signal levels. Calculate the maximum bit rate. [Model]

1. Bit rate = 2 x B x log2L = 2 x 6000 x log24
2. = 2 x 6000 x 2 = **24,000 bps = 24 kbps**.

A36. A noiseless channel with a bandwidth of 8000 Hz transmits a signal with eight signal levels. Calculate the maximum bit rate. [Model]

1. Bit rate = 2 x 8000 x log28 = 2 x 8000 x 3
2. = **48,000 bps = 48 kbps**.

A37. A noiseless channel has a bandwidth of 3000 Hz and transmits a signal with four levels. Calculate the maximum bit rate. [PYQ 2024-25 backlog]

1. Bit rate = 2 x 3000 x log24 = 2 x 3000 x 2
2. = **12,000 bps = 12 kbps**.

A38. A channel has B = 4 kHz and a signal-to-noise ratio of 30 dB. Determine the maximum information rate for 4-level encoding. [Model]

1. SNR = 10^(30/10) = 1000.
2. Shannon: C = 4000 x log2(1001) = 4000 x 9.967 = **39.87 kbps**.
3. Nyquist with 4 levels: 2 x 4000 x 2 = 16 kbps, which is inside the Shannon limit, so 4 levels are safe but far from optimal.

A39. A channel has a bandwidth of 10 MHz. Determine the capacity for a signal-to-noise ratio of 60 dB. [Model]

1. SNR = 10^6.
2. C = 10^7 x log2(1 + 10^6) = 10^7 x 19.93.
3. **C = 199.3 Mbps, about 200 Mbps.**

A40. A telephone line has a bandwidth of 5000 Hz and a signal-to-noise ratio of 3262. Calculate the channel capacity. [Model]

1. C = 5000 x log2(3263) = 5000 x 11.672.
2. **C = 58,360 bps, about 58.36 kbps.**

A41. A digital signal is to permit 160 kbps over a bandwidth of 20 kHz. Find the number of levels and the required signal-to-noise ratio. [Model]

1. Nyquist: 160,000 = 2 x 20,000 x log2L, so log2L = 4 and **L = 16 levels**.
2. Shannon: 160,000 = 20,000 x log2(1 + SNR), so log2(1 + SNR) = 8.
3. 1 + SNR = 256, so **SNR = 255** and SNRdB = 10 log10255 = **24.07 dB**.

A42. A complex low-pass signal has a bandwidth of 200 kHz. What is the minimum sampling rate? [Model]

1. For a low-pass signal the band runs from 0 to fmax, so fmax = 200 kHz.
2. fs = 2 x 200 kHz = **400,000 samples per second**.

A43. A complex bandpass signal has a bandwidth of 200 kHz. What is the minimum sampling rate? [PYQ 2024-25 backlog]

1. For a **bandpass** signal the band is centred on a carrier, and the sampling theorem applies to the bandwidth, not to the highest frequency.
2. fs = 2 x bandwidth = 2 x 200 kHz.
3. **fs = 400,000 samples per second.**

A44. If a periodic signal is decomposed into four sine waves with frequencies of 100, 300, 500 and 700 Hz, what is its bandwidth? [Model]

1. B = fhighest - flowest = 700 - 100.
2. **B = 600 Hz.**
3. The spectrum has four spikes, at 100, 300, 500 and 700 Hz.

A45. The energy strength at point 2 is 1/50 of that at point 1. Find the attenuation in dB. [Model]

1. dB = 10 log10(1/50) = 10 x (-1.699).
2. **Attenuation = -16.99 dB, about -17 dB.**

A46. A cable has a loss of -0.35 dB/km and the signal at the beginning has a power of 3 mW. What is the power at 6 km? [Model]

1. Loss over 6 km = 6 x (-0.35) = **-2.1 dB**.
2. -2.1 = 10 log10(P2/3), so log10(P2/3) = -0.21.
3. P2/3 = 10^(-0.21) = 0.6166.
4. **P2 = 1.85 mW.**

A47. What is crosstalk? [Model]

- Crosstalk is the **coupling of an unwanted signal** from one channel or wire pair into a neighbouring one, by electromagnetic induction.
- It is a form of **induced noise**.
- Reduced by twisting the pairs, shielding, and keeping pairs apart.

A48. Explain block coding and its use. [PYQ 2024-25 backlog]

- In block coding the message is divided into **k-bit datawords**; r redundant bits are added to make **n = k + r bit codewords**, using a substitution table.
- **Use 1**: error detection - most corrupted codewords are invalid and are discarded.
- **Use 2**: it guarantees signal transitions. **4B/5B** replaces each 4-bit group by a 5-bit code chosen to avoid long runs of 0s, so the line stays synchronised; 8B/10B does the same for gigabit links.
- Cost: the signal rate rises by the ratio n/k - 25 per cent for 4B/5B.

A49. Explain Internet standards and the maturity levels of a Request for Comment (RFC). [PYQ 2024-25 backlog]

- An **Internet standard** is a thoroughly tested specification that all Internet systems must follow; the documents are called **RFCs**.
- **Maturity levels**: Internet draft, then **proposed standard** (stable, under review), then **draft standard** (well tested, at least two independent implementations), then **Internet standard** (mature and widely used).
- **Requirement levels** of an RFC: required, recommended, elective, limited use, not recommended.
- Side tracks: historic, experimental, informational.

A50. How are OSI and ISO related to each other? [Model]

- **ISO** is the organisation - a multinational body founded in 1947 that promotes worldwide agreement on standards.
- **OSI** is a standard *produced by* ISO: the Open Systems Interconnection reference model, introduced in the late 1970s.
- One sentence: ISO is the body, OSI is its model.

A51. Distinguish between a port address, a logical address and a physical address. [Model; PYQ 2023]

Address	Layer and size	Purpose
Physical (MAC)	Data link, 48 bits in the NIC	Deliver a frame from node to node on one network
Logical (IP)	Network, 32 bits (IPv4)	Deliver a packet from source host to destination host across networks
Port	Transport, 16 bits	Deliver a message to the right process on the host

A52. What is the difference between network layer delivery and transport layer delivery? [Model]

Point	Network layer	Transport layer
Delivery scope	Source to destination host	Process to process
Address used	Logical (IP) address	Port address
Unit	Individual packet, handled independently	The entire message, segmented and reassembled
Guarantee	Best effort - may lose, reorder or duplicate	TCP adds reliability, ordering and flow control

Section B | 7-mark questions

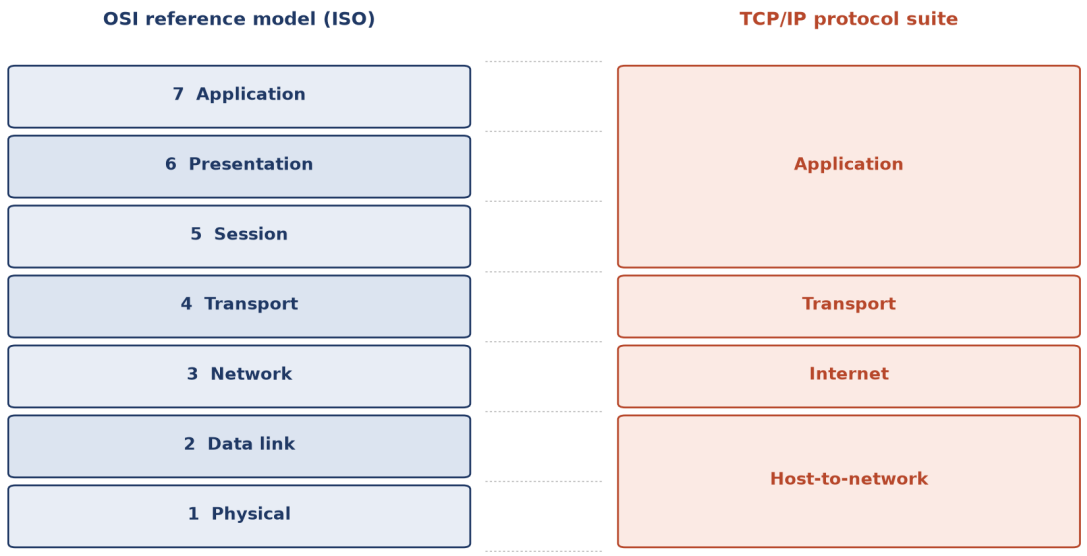
Write the heading points first, then expand each one. The bullet structure below is the marking scheme: one bullet is worth roughly half a mark to a mark.

B1. Explain the functionalities of the seven-layer OSI reference model with a neat diagram. Discuss the significance of the layered architecture. [PYQ 2024-25; PYQ 2022; PYQ 2023; PYQ backlog; Assignment 1]

Introduction. ISO, the International Organization for Standardization, produced the Open Systems Interconnection model in the late 1970s to cover all aspects of network communication. It divides the whole task into seven ordered layers.

1. **Physical** - physical transmission of bits: mechanical and electrical specifications of the interface and medium, representation of bits, data rate, bit synchronisation, line configuration, topology, transmission mode.
2. **Data link** - moves *frames* from one node to the next: framing, physical (MAC) addressing, flow control, error control, access control. It makes the physical layer appear error-free to the network layer.
3. **Network** - source-to-destination delivery of a *packet* across networks: logical (IP) addressing and routing.
4. **Transport** - process-to-process delivery of the *entire message*: port addressing, segmentation and reassembly, connection control, end-to-end flow and error control.
5. **Session** - establishes, manages and terminates sessions: dialog control and synchronisation by checkpoints.
6. **Presentation** - syntax and semantics: translation, encryption, compression.
7. **Application** - services to the user: file transfer and access management, mail services, directory services.

Mnemonic (layer 7 down to 1) - **A**ll **P**eople **S**eem **T**o **N**eed **D**ata **P**rocessing: Application, Presentation, Session, Transport, Network, Data link, Physical.



One TCP/IP layer may cover several OSI layers; the two models describe the same work at different granularity.

Significance of layering. Each layer has one well-defined job, so the design is modular and a layer can be redesigned without touching its neighbours. Faults can be isolated to one layer, products from different vendors interoperate because each layer's job is fixed by the standard, and the same transport layer can run over any physical medium. The cost is header overhead at every layer.

B2. What is the TCP/IP protocol suite? Explain its layers and protocols, and compare it with the OSI model. [PYQ 2022; PYQ 2023; Model]

Introduction. TCP/IP (Transmission Control Protocol / Internet Protocol) is a hierarchical suite of interactive modules, each providing a specific function. It is the model on which the Internet actually runs.

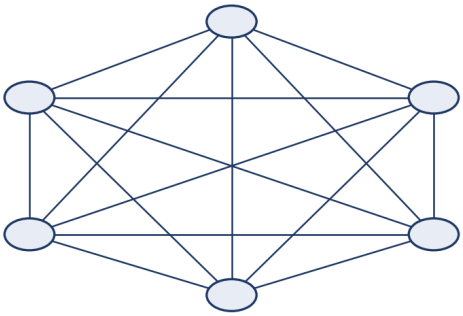
1. **Host-to-network** (OSI physical + data link) - transmits data between two devices on the same network.
2. **Internet** (OSI network) - host-to-host delivery using **IP** (datagrams, connectionless, best effort), **ARP** (IP to physical address), **RARP** (physical to IP), **ICMP** (error and query messages) and **IGMP** (group transmission).
3. **Transport** (OSI transport) - process-to-process delivery using **TCP** (connection-oriented, reliable) and **UDP** (connectionless, unreliable, fast).
4. **Application** (OSI session + presentation + application) - HTTP, SMTP, FTP, DNS, Telnet.

Point	OSI	TCP/IP
Layers	7	4 (5 in the Forouzan treatment)
Order of definition	Model first, protocols later	Protocols first, model describes them
Nature	General reference model, never fully implemented	Practical suite, actually deployed
Session and presentation	Separate layers	Merged into the application layer
Network layer service	Connection-oriented and connectionless	Connectionless only; reliability moved to transport

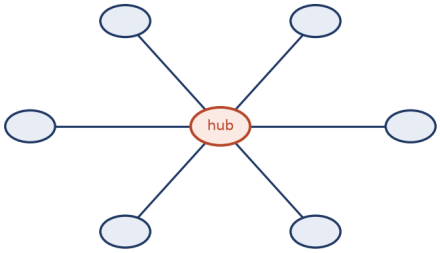
B3. Define topology. Explain the four basic topologies with their relative advantages and disadvantages. How many links does a fully connected mesh with 5 nodes need? [PYQ 2024-25; PYQ 2023; Model]

Definition. The topology of a network is the geometric representation of the relationship of all the links and linking devices, called nodes, to one another.

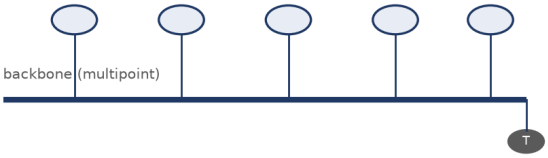
Mesh: n(n-1)/2 links, n(n-1) ports



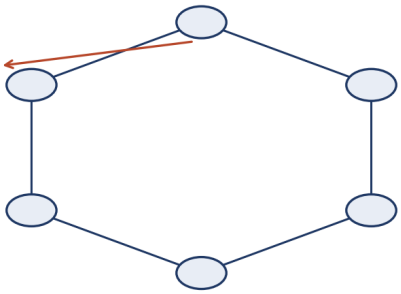
Star: n links, one per device to the hub



Bus: taps on a single backbone cable



Ring: n links, each device repeats the signal



Topology	Links for n devices	Advantage	Disadvantage
Mesh	n(n-1)/2, ports n(n-1)	Least traffic, robust, private, easy fault isolation	Huge cabling and port cost
Star	n	Robust per link, easy fault identification	Hub is a single point of failure
Bus	1 backbone plus drop lines	Least cabling	Hard to reconnect, reflections at taps, backbone failure kills all

Topology	Links for n devices	Advantage	Disadvantage
Ring	n	Easy to add a device, repeater per node	Unidirectional traffic, a break disables the ring

Mesh with 5 nodes: links = n(n-1)/2 = 5 x 4 / 2 = **10 links** (and 5 x 4 = 20 ports). The same five devices would need only 5 links in a ring or a star, which is why mesh is reserved for high-traffic backbones.

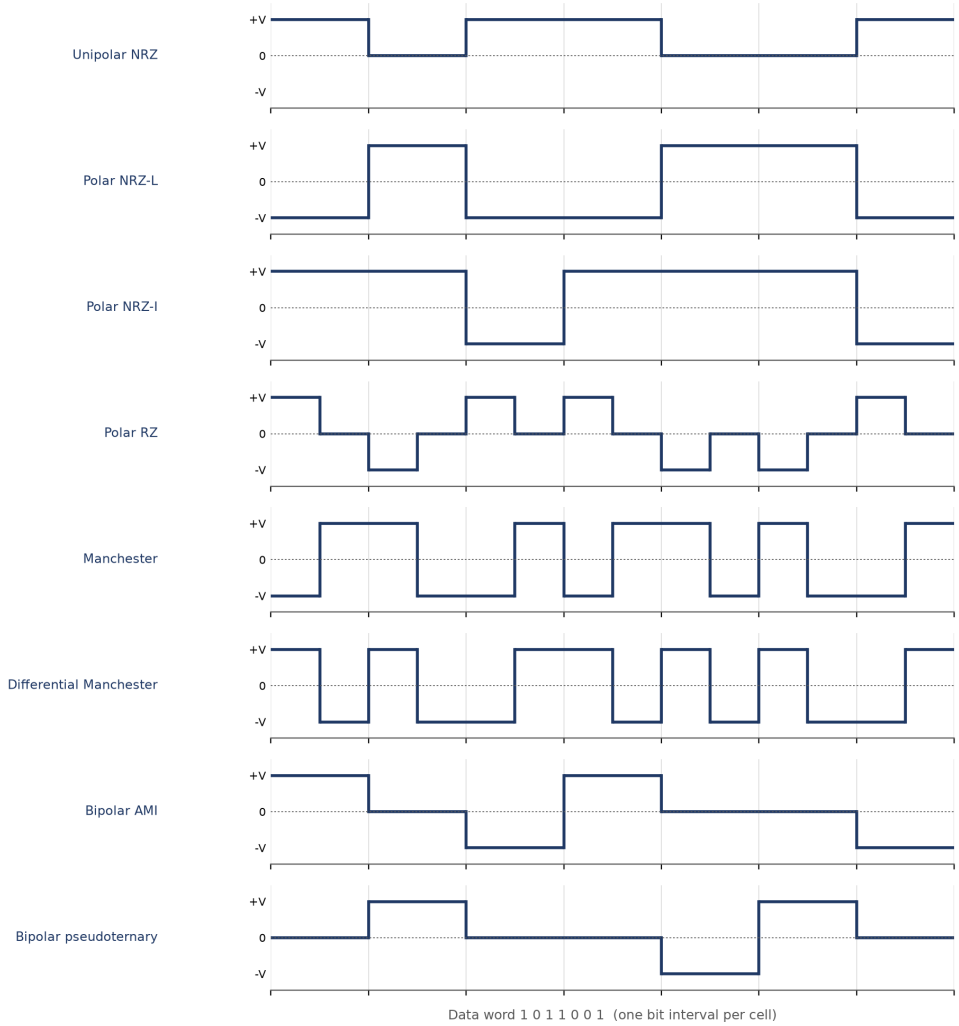
Effect on efficiency: mesh gives every pair a dedicated path, so delay is low and predictable; star centralises traffic in the hub, so the hub becomes the bottleneck; bus and ring share one medium, so throughput falls as stations are added.

B4. Encode the data word 1011001 using Unipolar NRZ, Polar NRZ-L, NRZ-I, RZ, Manchester, Differential Manchester, Bipolar AMI and Pseudoternary, and draw the waveforms. [Assignment 1; PYQ 2022; PYQ 2023; PYQ 2024-25; PYQ backlog]

Rules used (state them in the exam - they carry marks):

- **Unipolar NRZ**: 1 = +V, 0 = 0 V; no return to zero mid-bit.
- **Polar NRZ-L**: 0 = +V, 1 = -V (level carries the bit).
- **NRZ-I**: inversion at the start of the bit = 1, no inversion = 0; assume the line was low before the first bit.
- **RZ**: 1 = +V then 0; 0 = -V then 0 - every bit returns to zero mid-bit.
- **Manchester**: 0 = high-to-low, 1 = low-to-high at mid-bit.
- **Differential Manchester**: mid-bit transition always; 0 = transition at the start of the bit, 1 = no transition at the start.
- **AMI**: 0 = 0 V; 1 alternates +V, -V.
- **Pseudoternary**: 1 = 0 V; 0 alternates +V, -V.

Line coding of the data word 1011001



Reading the figure for 1011001. AMI sends +V, 0, -V, +V, 0, 0, -V; pseudoternary sends 0, +V, 0, 0, -V, +V, 0; NRZ-I inverts at bit 1, 3, 4 and 7 only. Manchester and differential Manchester show a transition inside every single bit, which is why they are self-clocking.

B5. Explain the line coding schemes and the criteria for choosing one, including MLT-3. [PYQ 2022; Notes]

Line coding converts a sequence of bits into a digital signal; the receiver decodes it back. Schemes are grouped as unipolar, polar, bipolar and multilevel/multitransition.

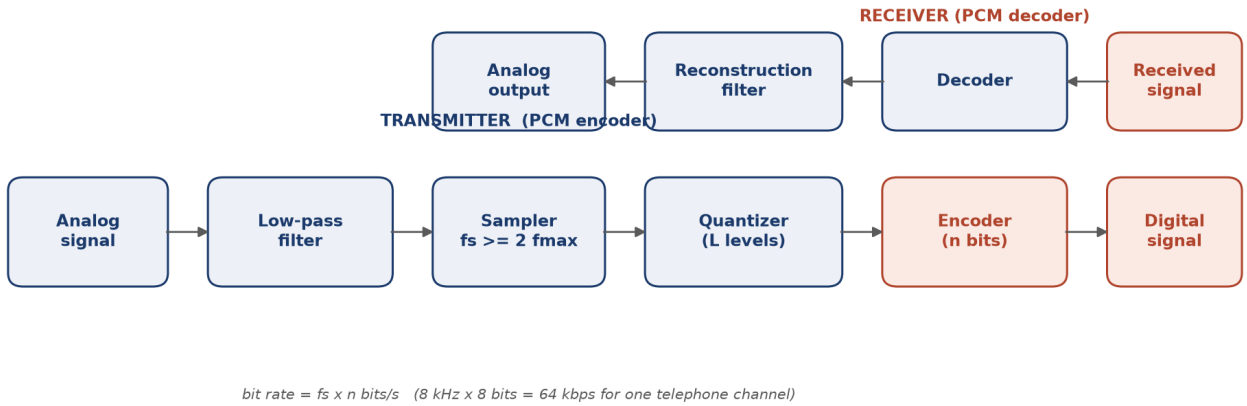
Scheme	Levels	DC free	Self-clocking	Notes
Unipolar NRZ	2	No	No	Simple, used only inside a box
Polar NRZ-L	2	No	No	Long runs skew the average
Polar NRZ-I	2	Partly	No	Long runs of 0s still a problem
Polar RZ	3	Yes	Yes	Twice the bandwidth
Manchester	2	Yes	Yes	10 Mbps Ethernet
Diff. Manchester	2	Yes	Yes	Token ring; tolerates inverted wires
Bipolar AMI	3	Yes	Partly	Alternation gives error detection
MLT-3	3	Yes	No	One third the bandwidth of NRZ

- **MLT-3** uses three levels (-V, 0, +V) and cycles through them. On a 1 the signal moves to the **next level in the cycle**; on a 0 it **stays** where it is. Because the level changes only on 1s, the highest fundamental frequency is one third of that of NRZ, so it needs about one third of the bandwidth - it is used in 100BASE-TX.
- **Criteria for a good line code**: no DC component (passes transformers), no baseline wandering, self-clocking, low signal rate for the data rate, some error-detection capability, noise immunity, low cost.

One-line summary to write: no scheme wins on every criterion; Manchester and RZ buy synchronisation with bandwidth, AMI buys error detection with three levels, and block codes such as 4B/5B and 8B/10B recover that bandwidth at high speed.

B6. Explain the different stages of pulse code modulation with a neat diagram, and compare it with delta modulation. [PYQ 2022; Model]

Why digitise. A digital signal is more robust to noise and can be regenerated, so analog data such as voice are converted to digital form before transmission.



bit rate = fs x n bits/s (8 kHz x 8 bits = 64 kbps for one telephone channel)

1 **Low-pass filter** - band-limits the signal so that fmax is known.

2. **Sampling** - the signal is measured every Ts, at fs at least 2 fmax. Methods: ideal (impulses), natural (pulse top follows the signal), flat-top (sample and hold, the practical method).
3. **Quantization** - each sample is rounded to the nearest of L levels; the rounding difference is quantization error, reduced by more levels or by non-uniform steps.
4. **Encoding** - each level becomes an n-bit code with L = 2^n; bit rate = fs x n.
5. **Receiver** - decoder, then a reconstruction low-pass filter reproduces an approximation of the original.

PCM versus DM. PCM sends several bits per sample and reproduces the signal accurately, at the cost of bandwidth and complexity. Delta modulation sends **one bit per sample** - only whether the signal rose or fell - so it needs far less bandwidth and simpler hardware, but it suffers slope-overload distortion when the signal changes faster than one step per sample, and granular noise when the signal is nearly constant.

B7. What are transmission impairments? Explain attenuation, distortion and noise, derive the expression for SNR in decibels, and discuss its significance. [Assignment 1; PYQ 2023]

Definition. Media are not perfect, so the signal at the end of the medium is not the signal that entered it. The three causes are attenuation, distortion and noise.

- **Attenuation** - loss of energy overcoming the resistance of the medium, so amplitude falls with distance. dB = 10 log10(P2/P1); negative means loss. Cable loss is quoted per km, e.g. -0.35 dB/km. Amplifiers or repeaters compensate.
- **Distortion** - change of shape. Each frequency component has its own propagation speed, so components arrive with different delays and different phases, and their sum is no longer the original shape. It affects composite signals only. Equalisation is the remedy.
- **Noise** - unwanted signal added on the way: **thermal** (random electron motion), **induced** (motors, appliances, crosstalk), **impulse** (lightning, power-line spikes).

Derivation of SNR in dB. SNR is the ratio of average signal power to average noise power: SNR = Ps/Pn. Taking logarithms compresses the huge range of values: SNRdB = 10 log10(Ps/Pn). For Ps = 10 mW and Pn = 1 microwatt, SNR = 10,000 and SNRdB = 40 dB.

Significance. SNR is the single number that sets the capacity of a real channel, through Shannon's law C = B log2(1 + SNR); it fixes how many signal levels can be distinguished; and it is what a link budget is built around. Every repeater, shield and twisted pair in a design exists to keep SNR high enough for the required bit error rate.

Memory hook - A-D-N: **A**ttenuation loses it, **D**istortion bends it, **N**oise adds to it.

B8. Explain the limits on data rate over a channel: the Nyquist bit rate and Shannon capacity, with worked examples. [Model]

Three limiting factors. The data rate depends on the bandwidth available, the number of signal levels, and the quality of the channel, that is the level of noise.

- **Noiseless channel (Nyquist)**: bit rate = 2 x B x log2L. The 2 comes from the sampling theorem - a signal of bandwidth B needs 2B samples per second, and each sample carries log2L bits. Example: B = 6000 Hz, L = 4 gives 24 kbps.
- **Noisy channel (Shannon)**: C = B log2(1 + SNR). Example: a telephone line with B = 3000 Hz and SNR = 3162 gives about 34.88 kbps - the origin of the 33.6 kbps modem.
- **Using them together**: Shannon fixes the ceiling, Nyquist says how many levels are needed to reach it. For 265 kbps over a noiseless 20 kHz channel, log2L = 6.625, so L must be about 98.7 - use 128 levels (280 kbps) or accept 64 levels (240 kbps).

Exam line: Shannon answers *is the target possible?*; Nyquist answers *what modulation is needed to reach it?*

B9. Explain the components, fundamental characteristics and data-flow modes of a data communication system. [Notes]

Definition. Data communication is the exchange of data between two devices through a transmission medium; telecommunication means communication at a distance and data means information in agreed form.

- **Components**: message, sender, receiver, transmission medium, protocol. Without the protocol two devices are connected but not communicating.
- **Characteristics**: delivery (right destination), accuracy (unchanged), timeliness (on time, in order), jitter (small variation in arrival time).

- **Simplex**: unidirectional, the whole capacity in one direction - keyboard, traditional monitor.
- **Half-duplex**: both ways but not at the same time, the transmitter takes the whole capacity - walkie-talkie.
- **Full-duplex**: both ways simultaneously, capacity divided between the directions - telephone.

Diagram to draw: three pairs of boxes with one arrow (simplex), two arrows marked one-at-a-time (half-duplex) and two arrows at once (full-duplex).

B10. Explain the network criteria, the types of connection, and compare LAN, MAN and WAN. [Notes]

- **Performance** - transit time, response time, throughput, delay; depends on users, medium and hardware.
- **Reliability** - frequency of failure, recovery time, robustness.
- **Security** - protection from unauthorised access, plus policies.
- **Point-to-point** - a dedicated link between exactly two devices, so the whole capacity belongs to that pair.
- **Multipoint** - one link shared by more than two devices, shared **spatially** (several at once) or **temporally** (take turns).

Parameter	LAN	MAN	WAN
Coverage	Room, building, campus	Town or city	Country, continent, world
Ownership	Private, one organisation	Government, ISP or large firm	Telecom operator or ISP
Speed	Highest (100 Mbps to Gbps)	High	Lower
Cost and delay	Low cost, low delay	Moderate	High cost, high delay
Medium	Ethernet, fiber, Wi-Fi	Fiber, microwave	Fiber, leased line, satellite
Example	College campus	Cable TV network	The Internet

B11. Explain protocols and standards, distinguish de facto from de jure standards, and name the standards organisations. [PYQ 2024-25 backlog; Notes]

- **Protocol** - rules governing communication: **syntax** (format), **semantics** (meaning), **timing** (when and how fast).
- **Standard** - guidelines for manufacturers and vendors that guarantee interconnectivity and compatibility and keep the market open and competitive.
- **De facto** - by fact or convention: not approved by a body but widely used because a manufacturer established it (the IBM PC, early modem standards).
- **De jure** - by law or regulation: recognised officially by a standards body (IEEE 802.3 Ethernet, ITU-T recommendations).
- **Organisations**: **ISO** (produced OSI), **ITU-T** (telecommunication), **IEEE** (Project 802 LAN standards), **ANSI** (US national body), **EIA** (interfaces such as RS-232), **IETF** (Internet standards and RFCs).
- **RFC maturity**: Internet draft, proposed standard, draft standard, Internet standard; requirement levels are required, recommended, elective, limited use, not recommended.

B12. Explain analog and digital signals: the sine wave and its parameters, the time and frequency domains, composite signals and bandwidth. [Notes]

- **Data versus signal**: analog data are continuous, digital data take discrete states; an analog signal has infinitely many levels, a digital signal only a limited number. Data communication commonly uses periodic analog signals and non-periodic digital signals.
- **Sine wave parameters**: amplitude (height, in volts or watts), frequency (cycles per second, Hz), phase (position relative to time zero, in degrees).
- **Relations**: f = 1/T, so 60 Hz has a period of 16.67 ms; wavelength = propagation speed / frequency, so 1 MHz in a cable at 2 x 10^8 m/s has a wavelength of 200 m.
- **Time domain** shows amplitude against time (the shape); **frequency domain** shows amplitude against frequency (the components) - spikes for a periodic signal, a curve for a non-periodic one.
- **Composite signal**: by Fourier analysis, any composite signal is a sum of sine waves of different frequency, amplitude and phase.
- **Bandwidth** = highest frequency - lowest frequency. Components at 100, 300, 500, 700 and 900 Hz give 800 Hz; without the 900 Hz component, 600 Hz.

Exam trap: bandwidth is the *width* of the band, not the centre frequency. A signal from 240 - 200 to 240 + 200 kHz has a bandwidth of 400 kHz.

B13. Explain the four levels of addressing used in TCP/IP with an example. [Model; Notes]

A message passes through several layers, and each layer needs a different identifier to do its job.

1. **Physical address** - 48-bit MAC address inside the NIC, written as 12 hexadecimal digits (07:01:02:01:2C:4B). Used by the data link layer for node-to-node delivery on one network.
2. **Logical address** - the IP address, 32 bits in IPv4 and 128 in IPv6. Used by the network layer for source-to-destination delivery across networks.
3. **Port address** - 16 bits, one decimal number such as 753. Used by the transport layer to reach the right process.
4. **Specific address** - a user-friendly address such as user@domain.com or www.example.com, translated by the application layer.

Worked chain: the browser resolves www.cgu-odisha.ac.in to an IP address, routers forward the packet on that IP address, the frame reaches the host on its MAC address, and the transport layer hands the data to the web server process on port 80.

B14. Compare analog and digital transmission, and justify why digital transmission is preferred today. [Notes]

Point	Analog transmission	Digital transmission
Signal	Continuous wave following the data	Discrete levels
Impairments	Added directly to the information	Removed by regeneration at each repeater
Noise	Amplifiers amplify noise too	Repeaters discard the noise
Error control	Difficult	Easy - only a few legal values
Integration	One service per medium	Voice, video and data in one stream
Bandwidth	Lower for one voice channel	Higher, but cheaper to run